Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Research Methods 3

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 4 NOV 2026

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About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

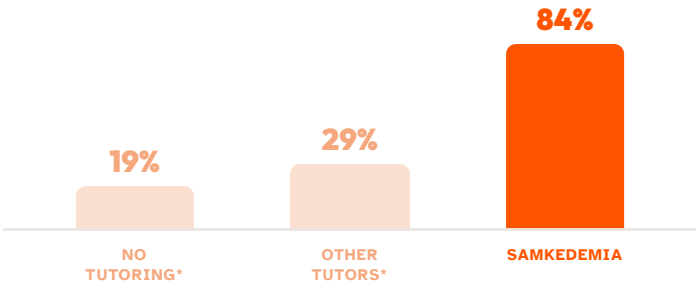
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

PROOF THAT IT WORKS.

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* BASED ON INTERNAL RESEARCH CONDUCTED IN 2026.

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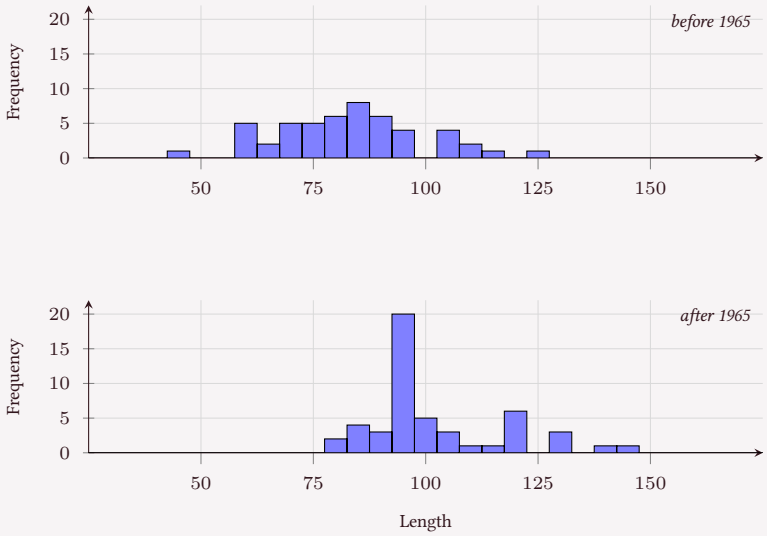
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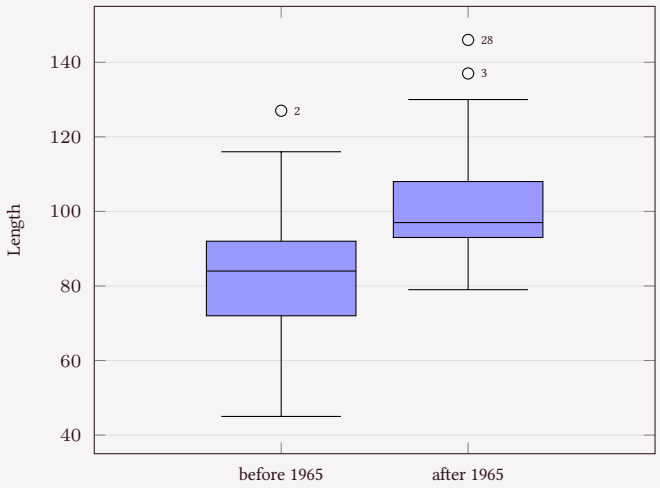
Chapter 1 - Hypothesis Testing

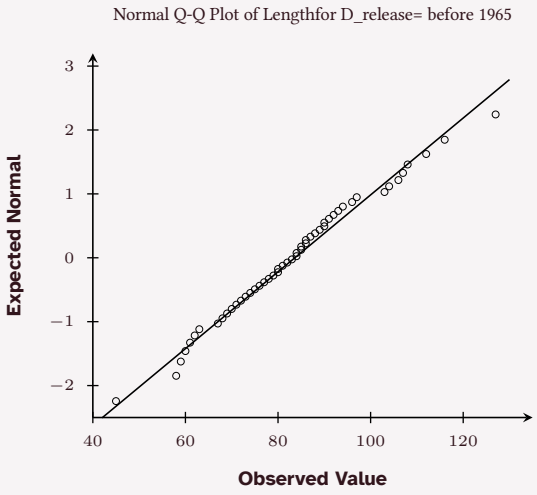
Question 1

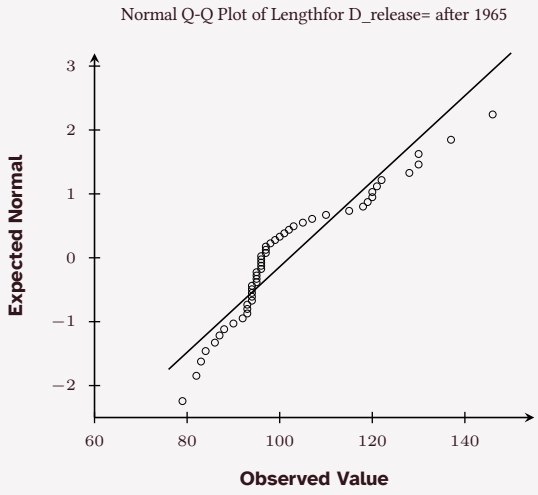
We consider data on a simple random sample of 100 movies (see file **movies.sav**). In a first analysis we want to test whether movies released before 1965 are on average longer than movies released after 1965. The variable **length** measures the length of a movie in minutes. The variable **D_release** indicates whether a movie was released before or after 1965.

Analysis with SPSS yields the following results:









Group Statistics

		N	Mean	Std. Deviation	Std. Error Mean
Length	before 1965	50	83.64	16.624	2.351
	after 1965	50	102.10	14.941	2.113

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Length	Equal variances assumed	.351	.555	-5.840	98	.000	-18.460	3.161	-24.733	-12.187
	Equal variances not assumed			-5.840	96.904	.000	-18.460	3.161	-24.734	-12.186

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a. Test whether movies released after 1965 are longer than movies released before 1965.

b. Discuss whether the assumptions of the independent-samples t-test are satisfied.

c. Explain what you can conclude from the "95% confidence interval of the difference" in the output of the independent samples t-test.

d. Indicate which test could be used to test whether the median length of movies released after 1965 is larger than the median length of movies released before 1965. Discuss whether the assumptions of this test are satisfied.

Tip: Independent-Samples t-test

The **independent-samples t-test** is a parametric test used to compare the population means of two independent samples.

$$H_0 : \mu_1 = \mu_2 \qquad H_A : \mu_1 \neq \mu_2$$

It comes with two assumptions:

- The two samples are independent (different subjects in each group, not the same subjects measured twice).
- $n_1, n_2 > 30$, so the CLT ensures the test statistic has an approximate t-distribution in each group. Otherwise, the variable must be Normally distributed within each group for the test statistic to have an exact t-distribution.

This course does not derive the test statistic by hand - it is only ever read off SPSS output.

Tip: Levene's Test

Levene's test checks whether the population variances of two or more groups are equal - it is used to check an assumption of the independent-samples t-test (and of ANOVA), not as a test of interest on its own.

$$H_0 : \sigma_1^2 = \sigma_2^2 \qquad H_A : \sigma_1^2 \neq \sigma_2^2$$

This course only ever reads F and $Sig.$ off SPSS output: if $Sig. \geq \alpha$, we do not reject equal variances; if $Sig. < \alpha$, we reject equal variances.

Solution 1: a. Testing whether movies released after 1965 are longer

We define the two variables as follows:

$$X_1 = \text{length (in minutes) of a movie released before 1965}$$

$$X_2 = \text{length (in minutes) of a movie released after 1965}$$

Step 1: Define the hypotheses.

We test whether the average length is larger for movies released after 1965:

$$H_0 : \mu_1 \geq \mu_2$$

$$H_A : \mu_1 < \mu_2$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the independent-samples t-test, we use the estimator T, the t-statistic that SPSS reports. Under H_0, T has a t-distribution, with the degrees of freedom df read off SPSS:

$$T \sim t_{df}$$

This is exact if the variable is Normally distributed in each group, and approximate if $n_1, n_2 > 30$ by the CLT.

Before reading the test value, we first check Levene's test for equality of variances, reported in the same Independent Samples Test table:

$$F = 0.351$$

$$Sig. = 0.555$$

Since $Sig. \geq \alpha$, hence we do not reject equal variances, and read the "Equal variances assumed" row in the Independent Samples Test table.

Step 3: Find the test value.
As mentioned above, we look at the "Equal variances assumed" row of the Independent Samples Test table. In the columns t and df, we see:

$$t = -5.840$$

$$df = 98$$

Step 4: Find the p-value.
SPSS reports a two-tailed significance in the same row:

$$Sig.\ (\text{2-tailed}) = 0.000$$

The mean difference between group 1 and group 2 is negative, so the test value t is negative as well, which matches the direction that H_A claims. We therefore divide the reported significance by two to obtain the one-tailed p-value:

$$p = \frac{0.000}{2} \approx 0.000$$

Step 5: Make the decision.
Since $p < \alpha$, we reject H_0.

Answer: We conclude with 95% confidence that movies released after 1965 are, on average,

longer than movies released before 1965.

Solution 1: b. Discuss whether the assumptions of the test are satisfied

The independent-samples t-test makes two assumptions:

- The two samples are independent.
- $n_1, n_2 > 30$, so the CLT ensures the test statistic has an approximate t-distribution in each group. Otherwise, the variable must be Normally distributed within each group.

Step 1: Check independence.
The before-1965 and after-1965 groups consist of different movies, not the same movies measured twice, so the two samples are independent. This assumption is satisfied.

Step 2: Check Normality/CLT.
Both groups have $n = 50 > 30$, so the CLT ensures the test statistic has an approximate t-distribution regardless of the shape of the population distribution. This assumption is satisfied. The boxplot does show a few outliers in each group (case 2 before 1965; cases 3 and 28 after 1965), but with $n = 50$ in each group the CLT still applies.

Answer: Both assumptions of the independent-samples t-test are satisfied.

Solution 1: c. Interpreting the 95% confidence interval of the difference

We read the 95% confidence interval of the difference from the same "Equal variances assumed" row of the Independent Samples Test table, in the "Lower" and "Upper" columns under 95%

Confidence Interval of the Difference:

$$[-24.733,\ -12.187]$$

We are 95% confident that the true population mean length of movies released before 1965 is, on average, between 24.733 and 12.187 minutes *shorter* than movies released after 1965. Since the entire interval lies below 0, this is consistent with the conclusion in part a: movies released after 1965 are on average longer.

Answer: With 95% confidence, movies released before 1965 are on average between 12.187 and 24.733 minutes shorter than movies released after 1965.

Tip: Mann-Whitney-Wilcoxon Test

The **Mann-Whitney-Wilcoxon test** is a non-parametric test used to compare the population medians of two independent samples, when the data is ordinal or non-normal quantitative.

$$H_0 : \text{median}_1 = \text{median}_2 \qquad H_A : \text{median}_1 \neq \text{median}_2$$

It comes with four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape (they can only differ by a shift in location).
- $n_1 + n_2 \geq 20$.

We prefer it over the independent-samples t-test when Normality/CLT fails, or the median is more meaningful than the mean.

Solution 1: d. Testing the median, and its assumptions

Step 1: Choose the test.

Movies released before 1965 and movies released after 1965 are two independent samples of different movies, not the same movies measured twice, and length is a quantitative variable. We are asked about the median, not the mean, so we use the non-parametric Mann-Whitney-Wilcoxon test to compare the two groups' medians, instead of the independent-samples t-test.

Step 2: Check the assumptions.

The Mann-Whitney-Wilcoxon test makes four assumptions:

Step 2.1: Check the variable type.

We are measuring length, which is a quantitative variable, so this assumption is satisfied.

Step 2.2: Check independence.

Movies released before 1965 and movies released after 1965 are different movies, not the same movies measured twice, so the two samples are independent. This assumption is satisfied.

Step 2.3: Check the same-shape assumption.

Looking at the two histograms, both distributions are unimodal and roughly bell-shaped, though the after-1965 group has a taller, narrower peak. Both boxplots also show outliers (case 2 before 1965; cases 3 and 28 after 1965). The two distributions look reasonably similar in shape, so this assumption is roughly satisfied.

Step 2.4: Check the sample-size assumption.

$n_1 + n_2 = 50 + 50 = 100 \geq 20$, so this assumption is satisfied.

Answer: We would use the Mann-Whitney-Wilcoxon test; all of its assumptions are satisfied (or roughly satisfied).

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Question 2

The file **movies.sav** also contains the rating (ranging from 1=very bad to 4=very good) of each movie from a certain critic. Some movie directors claim that this critic is unfair in that he assigns a negative rating (=rating lower than or equal to 2.5) to more than 65% of the movies.

The frequency distribution of movie ratings of the critic equals:

Rating

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1.0	8	8.0	8.0	8.0
	1.5	14	14.0	14.0	22.0
	2.0	19	19.0	19.0	41.0
	2.5	27	27.0	27.0	68.0
	3.0	25	25.0	25.0	93.0
	3.5	6	6.0	6.0	99.0
	4.0	1	1.0	1.0	100.0
Total		100	100.0	100.0	

a. Test whether more than 65% of the movies in the population receive a rating from the critic lower than or equal to 2.5.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Binomial Test

The **binomial test** is a non-parametric test used to test a population proportion π, for one category of a two-category variable in a single sample, against a hypothesized value π_0.

$$H_0 : \pi = \pi_0 \qquad H_A : \pi \neq \pi_0$$

It comes with three assumptions:

- Each observation is a Bernoulli trial: a yes/no outcome with the same probability π every

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time.

- The number of trials n is fixed in advance.
- The observations are independent of each other.

The estimator we use has an exact binomial distribution under H_0:

$$X \sim \text{Binom}(n,\ \pi_0) = \text{number of observations (out of } n\text{) in the category of interest}$$

Solution 2: a. Testing whether more than 65% of movies receive a negative rating

Step 1: Define the hypotheses.

Let π denote the population proportion of movies that receive a rating of 2.5 or lower from this critic. The movie directors claim this is more than 65%, so we test:

$$H_0 : \pi \leq 0.65$$

$$H_A : \pi > 0.65$$

$$\alpha = 0.05$$

Step 2: State the estimator.

We use the binomial test, with the estimator and its distribution under H_0:

$$X \sim \text{Binom}(n,\ \pi_0) = \text{number of movies (out of } n\text{) that receive a rating of 2.5 or lower}$$

Step 3: Find the test value.

Here $n = 100$ movies were sampled, and $\pi_0 = 0.65$ from H_0. In the Rating table, we read the Cumulative Percent column at the row for a rating of 2.5, which shows that 68.0% of the 100 movies received a rating of 2.5 or lower:

$$x = 68$$

Step 4: Find the p-value.

Since H_A says the proportion is higher than 0.65, we need the probability of a value at least as large as x:

$$p = P(X \geq 68 \mid H_0) = 1 - \texttt{binomcdf}(100,\ 0.65,\ 67) \approx 0.3029$$

Step 5: Make the decision.

Since $p \geq \alpha$, we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that more than 65% of the movies in the population receive a rating of 2.5 or lower from this critic - the population proportion could be 65% or lower.

Solution 2: b. Discuss whether the assumptions of the test are satisfied

The binomial test makes three assumptions:

- Each observation is a Bernoulli trial: a yes/no outcome with the same probability π every time.
- The number of trials n is fixed in advance.
- The observations are independent of each other.

Step 1: Check the Bernoulli trials.

Each movie's rating is a yes/no outcome (rating $\leq$ 2.5 or not), with the same probability π across movies, since every movie is rated by the same critic on the same scale. This assumption is satisfied.

Step 2: Check the fixed number of trials.

The number of movies sampled, $n = 100$, was fixed in advance, before observing the ratings.

This assumption is satisfied.

Step 3: Check independence.

The 100 movies in movies.sav are a simple random sample, so the observations are independent of each other. This assumption is satisfied.

Answer: All three assumptions of the binomial test are satisfied.

Question 3

The following cross-table shows for a random sample ($n = 183$) how many males and females would spend more than 500 euro when they buy a new smartphone.

gender * D_budget Crosstabulation

Count

		D_budget		
		less than 500 euro	more than 500 euro	Total
gender	male	46	29	75
	female	57	51	108
Total		103	80	183

a. Test whether the proportion of males who spend more than 500 euro on a new smartphone differs from the proportion of females who spend more than 500 euro.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Chi-Square Test for Proportions

The **chi-square test** is a non-parametric test used when the outcome is categorical (nominal), not quantitative. It is used to test whether the proportions of a categorical outcome differ between two or more independent samples.

$$H_0 : \pi_1 = \pi_2 = \dots \quad \text{(population proportions are equal)}$$

$$H_A : \text{at least one } \pi_i \text{ is different}$$

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Tip: Chi-Square Test: Mechanics

In order to do the chi-square test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O, with the groups as rows and the outcome categories as columns (or vice versa).
2. Compute the expected count for every cell, assuming H_0 is true:
$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$
3. Calculate the estimator:
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{(r-1)(k-1)}$$

with r the number of rows and k the number of columns in the table.

Solution 3: a. Testing whether the proportion who spend more than 500 euro differs between males and females

Step 1: Define the hypotheses.

Let $\pi_{more|i}$ be the proportion who would spend more than 500 euro on a new smartphone among people of gender i:

$$\pi_{more|M} = \text{proportion of males who would spend more than 500 euro}$$

$$\pi_{more|F} = \text{proportion of females who would spend more than 500 euro}$$

We test whether this proportion differs between males and females:

$$H_0 : \pi_{more|M} = \pi_{more|F}$$

$$H_A : \pi_{more|M} \neq \pi_{more|F}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Males and females are two independent groups of people, not the same people measured twice, and whether someone would spend more than 500 euro is a categorical variable (yes/no), not quantitative. This is exactly what the chi-square test for proportions is for: comparing a categorical outcome across two or more independent groups.

For the chi-square test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Under H_0, χ^2 has a chi-square distribution with $(r-1)(k-1)$ degrees of freedom.

Step 3: Find the test value.

None of the expected counts are given, so we compute all four of them:

$$E_{less|M} = \frac{75 \times 103}{183} \approx 42.2131 \quad E_{more|M} = \frac{75 \times 80}{183} \approx 32.7869$$

$$E_{less|F} = \frac{108 \times 103}{183} \approx 60.7869 \quad E_{more|F} = \frac{108 \times 80}{183} \approx 47.2131$$

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(46-42.2131)^2}{42.2131} + \frac{(29-32.7869)^2}{32.7869} + \frac{(57-60.7869)^2}{60.7869} + \frac{(51-47.2131)^2}{47.2131} \approx 1.3168$$

with $df = (2-1)(2-1) = 1$.

Step 4: Find the p-value.

The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 1.3168) = \chi^2\texttt{cdf}(\texttt{1.3168, 1E99, 1}) \approx 0.2512$$

Step 5: Make the decision.
The p-value is large, $p > 0.05$, so we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that the proportion of males who would spend more than 500 euro on a new smartphone differs from the proportion of females - the proportions could be equal.

Solution 3: b. Discuss whether the assumptions of the test are satisfied

The chi-square test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.
The four expected counts are 42.2131, 32.7869, 60.7869, and 47.2131, all larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the cells have an expected count smaller than 5.
None of the four expected counts are smaller than 5 (0% of the cells), so this assumption is satisfied too.

Answer: Both assumptions of the chi-square test are satisfied.

Question 4

The following cross-table shows for a random sample of 130 persons the relation between net household income and how much they would spend when they buy a new smartphone.

net household income * budget Crosstabulation

		budget					
		between 0 and 200 euro	between 200 and 500 euro	between 500 and 700 euro	between 700 and 900 euro	more than 900 euro	Total
less than 1000	Count	3	14	10	8	0	35
	Expected Count	7.8	12.1	6.5	5.7	3.0	35.0
between 1000 and 1500	Count	7	5	2	4	3	21
	Expected Count	4.7	7.3	3.9	3.4	1.8	21.0
between 1500 and 2000	Count	7	12	5	5	3	32
	Expected Count	7.1	11.1	5.9	5.2	2.7	32.0
between 2000 and 3000	Count	10	10	6	2	1	29
	Expected Count	6.5	10.0	5.4	4.7	2.5	29.0
more than 3000	Count	2	4	1	2	4	13
	Expected Count	2.9	4.5	2.4	2.1	1.1	13.0
Total	Count	29	45	24	21	11	130
	Expected Count	29.0	45.0	24.0	21.0	11.0	130.0

a. Test whether the variables **net household income** and **budget** are statistically dependent.

b. Discuss whether the assumptions of the test are satisfied, and indicate how you can proceed when the assumptions of the test are not satisfied.

Tip: Chi-Square Test of Independence

The **chi-square test** is a non-parametric test used to test whether two categorical (nominal) variables are statistically independent.

H_0 : the two variables are independent $\qquad$ H_A : the two variables are dependent

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

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It is mathematically the same test as the chi-square test for proportions: testing whether the proportions of one variable are equal across the categories of the other is the same as testing whether the two variables are independent.

Tip: Chi-Square Test: Mechanics

In order to do the chi-square test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O, with the groups as rows and the outcome categories as columns (or vice versa).
2. Compute the expected count for every cell, assuming H_0 is true:
$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$
3. Calculate the estimator:
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{(r-1)(k-1)}$$

with r the number of rows and k the number of columns in the table.

Solution 4: a. Testing whether net household income and budget are statistically dependent

Step 1: Define the hypotheses.

We test whether the two categorical variables are related:

$$H_0 : \text{net household income and budget are statistically independent}$$

$$H_A : \text{net household income and budget are statistically dependent}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Net household income and budget are both categorical variables (income classes and budget classes), and neither is a grouping/treatment variable - the question just asks whether the two are related. This is exactly what the chi-square test of independence is for.

For the chi-square test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Under H_0, χ^2 has a chi-square distribution with $(r-1)(k-1)$ degrees of freedom.

Step 3: Find the test value.

The expected counts are already shown in the crosstable above; computing them to more decimal places, e.g. for income less than 1000 and a budget between 0 and 200 euro:

$$E = \frac{35 \times 29}{130} \approx 7.8077$$

Repeating this for every cell gives:

income \ budget	0-200	200-500	500-700	700-900	>900
less than 1000	7.8077	12.1154	6.4615	5.6538	2.9615
between 1000 and 1500	4.6846	7.2692	3.8769	3.3923	1.7769
between 1500 and 2000	7.1385	11.0769	5.9077	5.1692	2.7077
between 2000 and 3000	6.4692	10.0385	5.3538	4.6846	2.4538
more than 3000	2.9000	4.5000	2.4000	2.1000	1.1000

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(3-7.8077)^2}{7.8077} + \cdots + \frac{(4-1.1)^2}{1.1} \approx 26.3015$$

with $df = (5-1)(5-1) = 16$.

Step 4: Find the p-value.
The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 26.3015) = \chi^2\texttt{cdf}(\texttt{26.3015, 1E99, 16}) \approx 0.0499$$

Step 5: Make the decision.
The p-value is small, $p < 0.05$, so we reject H_0.

Note that this p-value is very close to 0.05. With the rounded expected counts from the SPSS table, we would instead find:

$$\chi^2 \approx 26.2034 \qquad p \approx 0.0512$$

and we would not reject H_0.

Answer: We conclude with 95% confidence that net household income and budget are statistically dependent.

Solution 4: b. Discuss whether the assumptions of the test are satisfied, and how to proceed

The chi-square test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.
The smallest expected count is 1.1 (income more than 3000, budget more than 900 euro), which

is larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the cells have an expected count smaller than 5.
13 of the 25 expected counts are smaller than 5:

$$\frac{13}{25} = 52\% > 20\%$$

so this assumption is *not* satisfied, and we cannot trust the result of part a.

Step 3: How to proceed.
We can merge neighbouring categories, so that each cell contains more observations and the expected counts become larger. For example, we merge the two highest income classes (between 2000 and 3000, and more than 3000) and the two highest budget classes (between 700 and 900 euro, and more than 900 euro). This gives a 4×4 table in which the smallest expected count is 3.88 and only 2 of the 16 cells (12.5%) have an expected count smaller than 5, so both assumptions are satisfied. We then redo the chi-square test on this smaller table.

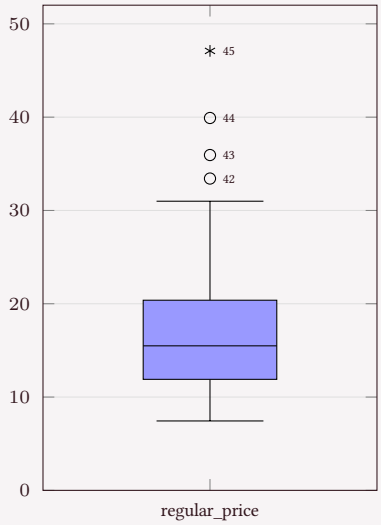
Answer: The first assumption is satisfied, but the second is not: 52% of the cells have an expected count smaller than 5. We can merge neighbouring income and budget classes until both assumptions are satisfied, and redo the test.

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Question 5

The file **price_wine.sav** contains the regular price and the promotion price for a random sample of 45 wines in a specialized wine store.

The following graph shows the distribution of the regular price in the sample:



We also compute descriptive statistics and the frequency distribution for the variable **regular_price**:

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
regular_price	45	7.44	47.10	17.7298	8.97385
Valid N (listwise)	45				

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regular_price

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	7.44	1	2.2	2.2	2.2
	7.81	1	2.2	2.2	4.4
	7.98	1	2.2	2.2	6.7
	8.18	1	2.2	2.2	8.9
	9.30	2	4.4	4.4	13.3
	9.92	1	2.2	2.2	15.6
	10.04	1	2.2	2.2	17.8
	10.29	1	2.2	2.2	20.0
	11.03	1	2.2	2.2	22.2
	11.53	1	2.2	2.2	24.4
	11.90	3	6.7	6.7	31.1
	12.39	2	4.4	4.4	35.6
	13.01	1	2.2	2.2	37.8
	13.14	1	2.2	2.2	40.0
	14.25	1	2.2	2.2	42.2
	14.58	1	2.2	2.2	44.4
	14.87	1	2.2	2.2	46.7
	15.49	2	4.4	4.4	51.1
	16.11	1	2.2	2.2	53.3
	16.36	1	2.2	2.2	55.6
	16.86	1	2.2	2.2	57.8
	17.10	1	2.2	2.2	60.0
	17.35	1	2.2	2.2	62.2
	17.48	1	2.2	2.2	64.4
	17.72	1	2.2	2.2	66.7
	17.97	1	2.2	2.2	68.9
	18.59	1	2.2	2.2	71.1
	20.20	1	2.2	2.2	73.3
	20.38	1	2.2	2.2	75.6
	22.19	1	2.2	2.2	77.8
	22.68	1	2.2	2.2	80.0
	22.93	1	2.2	2.2	82.2
	23.55	1	2.2	2.2	84.4
	28.51	1	2.2	2.2	86.7
	30.37	1	2.2	2.2	88.9
	30.99	1	2.2	2.2	91.1
	33.42	1	2.2	2.2	93.3
	35.94	1	2.2	2.2	95.6
	39.91	1	2.2	2.2	97.8
	47.10	1	2.2	2.2	100.0
Total		45	100.0	100.0	

a. Use a one-sample t-test to test whether the average regular price of wines in the store is larger than 14 euro. Discuss whether the assumptions of the test are satisfied.

b. Use a sign test to test whether the median price of regular wines in the store is larger than 14 euro. Discuss whether the assumptions of the test are satisfied.

c. Use a signed-rank test to test whether the median price of regular wines in the store is larger than 14 euro. Make the necessary computations in excel or SPSS. Discuss whether the assumptions of the test are satisfied.

Tip: One-Sample t-test

The **one-sample t-test** is a parametric test used to compare the population mean of one sample to a fixed value μ_0.

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

It comes with two assumptions:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the variable must be Normally distributed for the test statistic to have an exact t-distribution.

Tip: One-Sample t-test: Mechanics

In order to do the one-sample t-test, we use the estimator T. Here are the steps to find it, and its distribution under H_0:

1. Find the sample mean $\overline{x}$ and the sample standard deviation s of the n observations.
2. Calculate the estimator:

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}$$

Under H_0:

$$T \sim t_{n-1}$$

Solution 5: a. Testing whether the average regular price is larger than 14 euro (one-sample t-test)

Step 1: Define the hypotheses.

Let μ be the population mean regular price of the wines in the store. We test whether it is larger than 14 euro:

$$H_0 : \mu \leq 14$$

$$H_A : \mu > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the one-sample t-test, we use the estimator T:

$$T = \frac{\overline{X} - 14}{S/\sqrt{n}}$$

Under H_0, T has a t-distribution:

$$T \sim t_{n-1}$$

This is exact if the variable is Normally distributed, and approximate if $n > 30$ by the CLT.

Step 3: Find the test value.

In the Descriptive Statistics table, we read the sample size, the sample mean and the sample standard deviation of regular_price:

$$n = 45$$

$$\overline{x} = 17.7298$$

$$s = 8.97385$$

Plugging these into the estimator:

$$t = \frac{17.7298 - 14}{8.97385/\sqrt{45}} = \frac{3.7298}{1.3377} \approx 2.7881$$

with $df = n - 1 = 44$.

Step 4: Find the p-value.
Since H_A says the mean is larger, we need the probability of a value at least as large as t:

$$p = P(T \geq 2.7881) = \texttt{tcdf}(2.7881,\ \texttt{1E99},\ 44) \approx 0.0039$$

Step 5: Make the decision.
The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the one-sample t-test:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the variable must be Normally distributed for the test statistic to have an exact t-distribution.

The boxplot shows that regular_price is right-skewed, with a few outliers at the top, so the variable is not Normally distributed. But here $n = 45 > 30$, so the CLT still ensures that the test statistic has an approximate t-distribution, and the assumptions are satisfied.

Answer: We conclude with 95% confidence that the average regular price of the wines in the store is larger than 14 euro.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.
3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 5: b. Testing whether the median regular price is larger than 14 euro (sign test)

Step 1: Define the hypotheses.

We test whether the median regular price is larger than 14 euro:

$$H_0 : \text{Median}_{price} \leq 14$$

$$H_A : \text{Median}_{price} > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the sign test, we use the estimator B:

$$B = \text{number of observations where } price_i - 14 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.

None of the prices equal 14, so all $n = 45$ observations are kept. In the frequency table, the Cumulative Percent column shows that 40.0% of the wines, i.e. 18 of the 45, have a price of 13.14 or lower, and the next price is 14.25. So the other 27 wines have a price larger than 14:

$$b = 45 - 18 = 27$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 27) = 1 - \texttt{binomcdf}(45,\ 0.5,\ 26) \approx 0.1163$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The regular price is quantitative, so this assumption is satisfied.

Answer: We cannot conclude, with 95% confidence, that the median regular price of the wines in the store is larger than 14 euro - the median could be 14 euro or lower.

Tip: Wilcoxon Signed-Rank Test

The **Wilcoxon signed-rank test** is a non-parametric test used when data is ordinal or non-normal quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

We prefer it over the sign test, because it uses the size of each deviation, not just its sign.

Tip: Signed-Rank Test: Mechanics

In order to do the Wilcoxon signed-rank test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.
3. Rank $|x_i - med_0|$ from smallest to largest (the smallest gets rank 1, the next smallest rank 2, and so on).

   3.1 Ties get the average rank (e.g. if the 5th and 6th are tied, both get 5.5).
4. Calculate the estimator:

$$B = \text{sum of ranks for observations where } x_i - med_0 > 0$$

Under H_0:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Solution 5: c. Testing whether the median regular price is larger than 14 euro (signed-rank test)

Step 1: Define the hypotheses.

We test whether the median regular price is larger than 14 euro:

$$H_0 : \text{Median}_{price} \leq 14$$

$$H_A : \text{Median}_{price} > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the Wilcoxon signed-rank test, we use the estimator B:

$$B = \text{sum of ranks for observations where } price_i - 14 > 0$$

Under H_0, B is approximately Normal:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Step 3: Find the test value.

For each observation, we compute the deviation $price_i - 14$. None of the prices equal 14, so all $n = 45$ observations are kept. We then rank $|price_i - 14|$ from smallest to largest.

Several prices appear more than once (two wines of 15.49, two of 12.39, three of 11.90 and two of 9.30), so their $|price_i - 14|$ are tied, and each group is given the average of the ranks they would otherwise occupy. Repeating this for every observation gives:

price	$price_i - 14$	$\lvert price_i - 14\rvert$	rank	price	$price_i - 14$	$\lvert price_i - 14\rvert$	rank
14.25	0.25	0.25	1	17.97	3.97	3.97	24
14.58	0.58	0.58	2	9.92	−4.08	4.08	25
13.14	−0.86	0.86	3	18.59	4.59	4.59	26
14.87	0.87	0.87	4	9.30	−4.70	4.70	27.5
13.01	−0.99	0.99	5	9.30	−4.70	4.70	27.5
15.49	1.49	1.49	6.5	8.18	−5.82	5.82	29
15.49	1.49	1.49	6.5	7.98	−6.02	6.02	30
12.39	−1.61	1.61	8.5	7.81	−6.19	6.19	31
12.39	−1.61	1.61	8.5	20.20	6.20	6.20	32
11.90	−2.10	2.10	11	20.38	6.38	6.38	33
11.90	−2.10	2.10	11	7.44	−6.56	6.56	34
11.90	−2.10	2.10	11	22.19	8.19	8.19	35
16.11	2.11	2.11	13	22.68	8.68	8.68	36
16.36	2.36	2.36	14	22.93	8.93	8.93	37
11.53	−2.47	2.47	15	23.55	9.55	9.55	38
16.86	2.86	2.86	16	28.51	14.51	14.51	39
11.03	−2.97	2.97	17	30.37	16.37	16.37	40
17.10	3.10	3.10	18	30.99	16.99	16.99	41
17.35	3.35	3.35	19	33.42	19.42	19.42	42
17.48	3.48	3.48	20	35.94	21.94	21.94	43
10.29	−3.71	3.71	21	39.91	25.91	25.91	44
17.72	3.72	3.72	22	47.10	33.10	33.10	45
10.04	−3.96	3.96	23				

The observed sum of positive ranks:

$$b = 1 + 2 + 4 + 6.5 + 6.5 + 13 + 14 + 16 + 18 + 19 + 20 + 22 + 24 + 26$$

$$+ 32 + 33 + 35 + 36 + 37 + 38 + 39 + 40 + 41 + 42 + 43 + 44 + 45 = 697$$

With $n = 45$:

$$E(B) = \frac{(45)(46)}{4} = 517.5000$$

$$Var(B) = \frac{(45)(46)(91)}{24} = 7848.7500$$

$$SD(B) = \sqrt{7848.7500} = 88.5932$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 697) = \texttt{normalcdf}(697,\ \texttt{1E99},\ 517.5,\ 88.5932) \approx 0.0214$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the signed-rank test:

- The variable is quantitative or ordinal.
- $n > 20$.

The regular price is quantitative, so the first assumption is satisfied. Here $n = 45 > 20$, so the second assumption is satisfied too.

Answer: We conclude with 95% confidence that the median regular price of the wines in the store is larger than 14 euro.

Question 6

The file **price_wine.sav** contains both the regular price and the promotion price for a sample of 45 wines from a specialized wine store. The frequency distribution of the regular price minus the promotion price is displayed in the table below:

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regular price minus promotion price

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0.73	1	2.2	2.2	2.2
	0.86	1	2.2	2.2	4.4
	0.95	1	2.2	2.2	6.7
	1.00	2	4.4	4.4	11.1
	1.03	1	2.2	2.2	13.3
	1.14	1	2.2	2.2	15.6
	1.15	2	4.4	4.4	20.0
	1.18	1	2.2	2.2	22.2
	1.19	1	2.2	2.2	24.4
	1.24	1	2.2	2.2	26.7
	1.30	1	2.2	2.2	28.9
	1.31	1	2.2	2.2	31.1
	1.34	1	2.2	2.2	33.3
	1.36	1	2.2	2.2	35.6
	1.37	1	2.2	2.2	37.8
	1.39	1	2.2	2.2	40.0
	1.40	1	2.2	2.2	42.2
	1.44	1	2.2	2.2	44.4
	1.48	1	2.2	2.2	46.7
	1.50	2	4.4	4.4	51.1
	1.53	1	2.2	2.2	53.3
	1.56	1	2.2	2.2	55.6
	1.59	1	2.2	2.2	57.8
	1.60	1	2.2	2.2	60.0
	1.64	1	2.2	2.2	62.2
	1.64	1	2.2	2.2	64.4
	1.65	1	2.2	2.2	66.7
	1.66	2	4.4	4.4	71.1
	1.68	1	2.2	2.2	73.3
	1.70	1	2.2	2.2	75.6
	1.73	1	2.2	2.2	77.8
	1.81	1	2.2	2.2	80.0
	1.86	1	2.2	2.2	82.2
	1.98	1	2.2	2.2	84.4
	2.39	1	2.2	2.2	86.7
	2.43	1	2.2	2.2	88.9
	2.44	1	2.2	2.2	91.1
	2.49	1	2.2	2.2	93.3
	2.93	1	2.2	2.2	95.6
	3.87	1	2.2	2.2	97.8
	3.94	1	2.2	2.2	100.0
Total		45	100.0	100.0	

Conduct a sign test to test whether the median price reduction in the store for a wine in promo-

tion is larger than 1 euro. Compute the test statistic of the sign test from the information in the frequency distribution of regular price minus promotion price. Discuss whether the assumptions of the test are satisfied.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.

3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 6: Testing whether the median price reduction is larger than 1 euro (sign test)

Let X denote the price reduction of a wine in promotion, i.e. the regular price minus the promotion price.

Step 1: Define the hypotheses.

We test whether the median price reduction is larger than 1 euro:

$$H_0 : \text{Median}_X \leq 1$$

$$H_A : \text{Median}_X > 1$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the sign test, we use the estimator B:

$$B = \text{number of observations where } X_i - 1 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.

In the frequency table, two wines have a price reduction of exactly 1.00, so their deviation from 1 is 0 and they are dropped, leaving $n = 45 - 2 = 43$ observations.

The Cumulative Percent column shows that 11.1% of the wines, i.e. 5 of the 45, have a price reduction of 1.00 or lower. Two of those 5 are exactly 1.00, so 3 wines have a reduction smaller than 1, and all the others have a reduction larger than 1:

$$b = 43 - 3 = 40$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 40) = 1 - \texttt{binomcdf}(43,\ 0.5,\ 39) \approx 0.0000$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The price reduction is quantitative, so this assumption is satisfied.

Answer: We conclude with 95% confidence that the median price reduction in the store for a wine in promotion is larger than 1 euro.

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Chapter 2 - Regression

Chapter 3 - Time Series

Exam: 2022 January

Exam: 2022 August

Exam: 2024 Example 1

Exam: 2024 Example 2

Exam: 2024 January

Exam: 2024 August

Exam: 2025 January

Exam: 2025 August

Exam: 2026 Example

Exam: 2026 January

Exam: Unknown 1

Exam: Unknown 2

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